

R Notebook

```
library(tidyverse) # For data manipulation and visualization
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(vegan) # For diversity analyses
```

```
## Loading required package: permute
```

```
library(phyloseq) # For microbiome data analysis
library(ggplot2)  # For enhanced visualizations
library(microbiome)
```

```
##
## microbiome R package (microbiome.github.com)
##
##
## Copyright (C) 2011-2022 Leo Lahti,
## Sudarshan Shetty et al. <microbiome.github.io>
##
##
## Attaching package: 'microbiome'
##
## The following object is masked from 'package:vegan':
##
##     diversity
##
## The following object is masked from 'package:ggplot2':
##
##     alpha
##
## The following object is masked from 'package:base':
##
##     transform
```

```
library(dplyr)
library(tidyr)
library(patchwork)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a",
                        full.names = TRUE)
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m")

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
pond_samples = c("s9", "s10", "s19", "s20")

# Create a metadata dataframe to merge with taxon data
sample_metadata = seq_metrics %>%
  select(sample_name, water_control, infection_class) %>%
  mutate(
    # Library prep: MSSPE if ends with "_M"
    library_prep = if_else(str_ends(sample_name, "_M"), "MSSPE", "NGS"),

    # Remove the _M suffix to get the core sample identifier
    core = str_remove(sample_name, "_M$"),
    # Detect experiment 3 first (Cxxx_LPx)
    is_exp3 = str_detect(core, "C\\d+.*LP\\d+"),

    # Base: first sample ID (e.g. S10, PC1)
    base = str_extract(core, "[A-Za-z]+\\d+"),

    # Append -2 or -3 only if not exp3
    suffix = case_when(
      !is_exp3 & str_detect(core, "_2$") ~ "-2",
      !is_exp3 & str_detect(core, "_3$") ~ "-3",
      TRUE ~ ""
    ),

    base_sample = tolower(paste0(base, suffix)),

    # group slurry vs environmental samples
```

```

sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry"),

experiment = case_when(
  is_exp3 ~ 3,
  str_detect(core, "_3$") ~ 2,
  TRUE ~ 1
)
) %>%
select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  #filter(experiment == 2 | experiment == 3) %>% # filter in specific experiments
  filter(water_control == "No") %>% # filter out Negative Controls
  filter(!base_sample %in% c("pc1", "pc3")) %>% # filter out zymo fecal controls
  filter(infection_class != "Definite") # filter out positive controls

```

Positive Inclusion Criteria

```

species_data_filtered = all_data_species %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >=3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)
  #filter(sample_type == "slurry") # filtering by sample type

```

Taxonomic Composition

```

# Create a taxa table
taxa_table <- species_data_filtered %>%
  select(tax_id, tax_level, name, genus_tax_id, category, library_prep, nt_rpm, nt_count, base_sample, )
  distinct() %>% # only unique taxa
  filter(!is.na(category)) %>% # filter out species with no aligned category
  filter(category %in% c("archaea", "bacteria", "eukaryota", "viruses")) # because some blank categories are present

phylum_composition_overall <- taxa_table %>%
  group_by(category) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

# Compare library prep at different taxonomic levels (e.g., phylum level)
phylum_composition <- taxa_table %>%
  group_by(category, library_prep) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(library_prep) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

```

`summarise()` has grouped output by 'category'. You can override using the
`.groups` argument.

```

sampletype_composition <- taxa_table %>%
  mutate(
    sample_type = recode(sample_type, `environmental` = "wastewater")
  ) %>%
  group_by(category, sample_type) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(sample_type) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

```

`summarise()` has grouped output by 'category'. You can override using the
`.groups` argument.

```

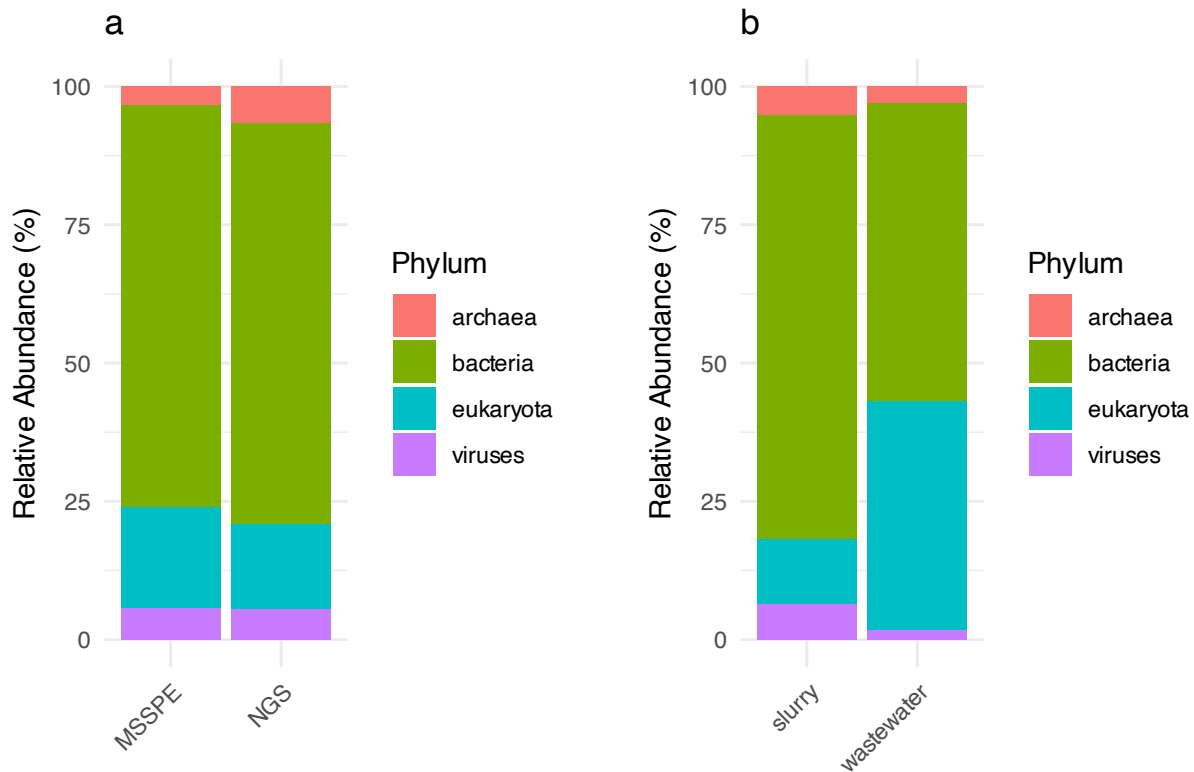
# Visualize composition by library prep method
tax_comp_libprep = ggplot(phylum_composition, aes(x = library_prep, y = relative_abundance, fill = category)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "a",
       x = "",
       y = "Relative Abundance (%)",
       fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

tax_comp_sampletype = ggplot(sampletype_composition, aes(x = sample_type, y = relative_abundance, fill = category)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "b",
       x = "",
       y = "Relative Abundance (%)",
       fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

combined_tax_plots = (tax_comp_libprep | plot_spacer() | tax_comp_sampletype) +
  plot_layout(ncol = 3, widths = c(2, 0.2, 2))

combined_tax_plots

```



Diversity

```
cat("\\captionsetup[labelformat=empty]")
```

```
# Function to compute diversity comparison table
compute_diversity_results <- function(df) {
  species_agg <- df %>%
    group_by(sample_name, tax_id) %>%
    summarise(nt_count = sum(nt_count, na.rm = TRUE), .groups = "drop")

  abundance_matrix <- species_agg %>%
    pivot_wider(names_from = sample_name, values_from = nt_count, values_fill = 0)

  abund_mat <- as.matrix(abundance_matrix[, -1])
  rownames(abund_mat) <- abundance_matrix$tax_id

  diversity_tbl <- data.frame(
    sample_name = colnames(abund_mat),
    Shannon = apply(abund_mat, 2, vegan::diversity, index = "shannon"),
    Simpson = apply(abund_mat, 2, vegan::diversity, index = "simpson"),
    Richness = apply(abund_mat, 2, function(x) sum(x > 0))
  ) %>%
    left_join(df %>%
      select(sample_name, library_prep, base_sample, sample_type) %>%
      distinct(),
      by = "sample_name")

  diversity_paired <- diversity_tbl %>%
    pivot_wider(
```

```

    id_cols = base_sample,
    names_from = library_prep,
    values_from = c(Shannon, Simpson, Richness)
  )

metrics <- c("Shannon", "Simpson", "Richness")

results <- lapply(metrics, function(metric) {
  ngs <- diversity_paired[[paste0(metric, "_NGS")]]
  msspe <- diversity_paired[[paste0(metric, "_MSSPE")]]

  test <- wilcox.test(ngs, msspe, paired = TRUE)

  data.frame(
    metric = metric,
    NGS = mean(ngs, na.rm = TRUE),
    MSSPE = mean(msspe, na.rm = TRUE),
    `p-value` = test$p.value
  )
})

bind_rows(results)
}

# ---- Create three datasets ----
species_slurry <- species_data_filtered %>% filter(sample_type == "slurry")
species_env <- species_data_filtered %>% filter(sample_type == "environmental")

# ---- Compute results ----
tbl_overall <- compute_diversity_results(species_data_filtered) %>% mutate(Type = "Overall")
tbl_slurry <- compute_diversity_results(species_slurry) %>% mutate(Type = "Slurry")
tbl_env <- compute_diversity_results(species_env) %>% mutate(Type = "Wastewater")

# ---- Combine ----
results_all <- bind_rows(tbl_slurry, tbl_env, tbl_overall) %>%
  rename("Diversity Measure" = metric)

# ---- Final kable ----
results_all %>%
  kable(
    format = "latex",
    caption = "Comparison of diversity metrics between library preparation methods by sample type",
    digits = 3,
    booktabs = TRUE
  ) %>%
  kable_styling(full_width = TRUE, position = "center") %>%
  collapse_rows(columns = 1, valign = "top")

# 1. Aggregate counts per tax_id per sample
species_agg <- species_data_filtered %>%
  group_by(sample_name, tax_id) %>%
  summarise(nt_count = sum(nt_count, na.rm = TRUE), .groups = "drop")

```

Comparison of diversity metrics between library preparation methods by sample type

Diversity Measure	NGS	MSSPE	p.value	Type
Shannon	5.470	5.641	0.055	Slurry
Simpson	0.980	0.982	0.073	Slurry
Richness	5179.600	6696.435	0.010	Slurry
Shannon	5.645	6.030	0.250	Wastewater
Simpson	0.979	0.984	0.250	Wastewater
Richness	8573.000	8632.000	1.000	Wastewater
Shannon	5.507	5.699	0.007	Overall
Simpson	0.980	0.983	0.016	Overall
Richness	5894.000	6983.185	0.018	Overall

```

# 2. Create abundance matrix (taxa x samples)
abundance_matrix <- species_agg %>%
  pivot_wider(
    names_from = sample_name,
    values_from = nt_count,
    values_fill = 0
  )

# Convert to numeric matrix and set rownames
abund_mat <- as.matrix(abundance_matrix[, -1])
rownames(abund_mat) <- abundance_matrix$tax_id

# 3. Compute diversity metrics per sample
diversity_tbl <- data.frame(
  sample_name = colnames(abund_mat),
  Shannon = apply(abund_mat, 2, vegan::diversity, index = "shannon"),
  Simpson = apply(abund_mat, 2, vegan::diversity, index = "simpson"),
  Richness = apply(abund_mat, 2, function(x) sum(x > 0))
)

# 4. Add library prep info
diversity_tbl <- diversity_tbl %>%
  left_join(species_data_filtered %>%
    select(sample_name, library_prep, base_sample, sample_type) %>%
    distinct(), by = "sample_name")

# Compare alpha diversity between library prep methods
shannon_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Shannon, color = library_prep)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "a",
    x = "",
    y = "Shannon Diversity Index") +
  theme(legend.position = "none")

simpson_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Simpson, color = library_prep)) +

```

```

geom_boxplot() +
theme_minimal() +
labs(title = "b",
      x = "",
      y = "Simpson Diversity Index") +
theme(legend.position = "none")

rich_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Richness, color = library_prep)) +
geom_boxplot() +
theme_minimal() +
labs(title = "c",
      x = "",
      y = "Species Richness") +
theme(legend.position = "none")

horizontal_space <- plot_spacer()
combined_diversity_plots = (shannon_libprep_plot | horizontal_space | simpson_libprep_plot | horizontal.
plot_layout(ncol = 5, widths = c(2, 0.2, 2, 0.2, 2))

```

```

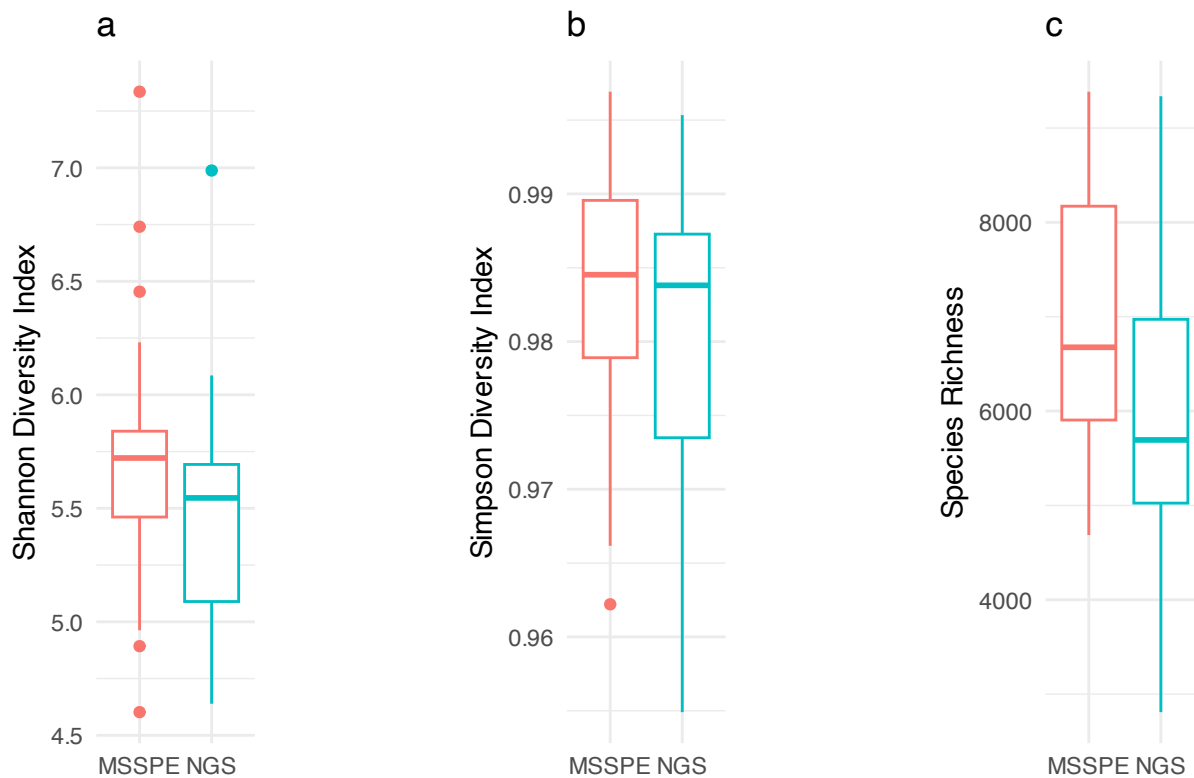
## $ncol
## [1] 5
##
## $nrow
## list()
## attr("class")
## [1] "waiver"
##
## $byrow
## list()
## attr("class")
## [1] "waiver"
##
## $widths
## [1] 2.0 0.2 2.0 0.2 2.0
##
## $heights
## list()
## attr("class")
## [1] "waiver"
##
## $guides
## list()
## attr("class")
## [1] "waiver"
##
## $tag_level
## list()
## attr("class")
## [1] "waiver"
##
## $axes
## list()
## attr("class")

```



```
## [1] "waiver"
##
## $axis_titles
## list()
## attr("class")
## [1] "waiver"
##
## $design
## list()
## attr("class")
## [1] "waiver"
##
## attr("class")
## [1] "plot_layout"
```

```
combined_diversity_plots
```



Top Viral Genera

```
# Extract viral category data
virus_data <- species_data_filtered %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false") %>% # filter out phages as marked by CZID
  filter(genus_tax_id != -200)

# Add genus columns to the dataframe
genus_data <- get_genus_info_batch(virus_data$tax_id, batch_size = 100)
```

```
## Processing 687 unique tax IDs out of 3747 total
```

```

## Processing batch 1 of 7 - IDs: 12239 to 3060732
## Processing batch 2 of 7 - IDs: 47681 to 3080759
## Processing batch 3 of 7 - IDs: 12242 to 3060074
## Processing batch 4 of 7 - IDs: 10288 to 3118160
## Processing batch 5 of 7 - IDs: 12196 to 3060737
## Processing batch 6 of 7 - IDs: 12205 to 3060826
## Processing batch 7 of 7 - IDs: 10665 to 3052721
## Successfully retrieved genus information for 2701 out of 3747 valid tax IDs

# Combine with original dataframe
virus_data <- cbind(virus_data, genus_data) %>%
  filter(!genus_name %in% c("Caudoviricetes sp.", "Crassvirales sp.", "Mitoviridae sp.", "uncultured human

cat("\\captionsetup[labelformat=empty]")

# get table of only unique species
taxa_table_genus <- virus_data %>%
  select(tax_id, tax_level, name, genus_tax_id, genus_name, category, library_prep, nt_rpm, nt_count, b
  distinct() %>% # only unique taxa
  filter(!is.na(genus_name)) # filter out species with no aligned category

# Calculate viral relative abundance of each viral genus (among all viruses)
genus_composition <- taxa_table_genus %>%
  #group_by(genus_name, library_prep) %>%
  group_by(genus_name) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  #group_by(library_prep) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

# Calculate mean nt_rpm for each taxon across samples
genus_summary <- virus_data %>%
  group_by(genus_name, library_prep) %>%
  filter(!genus_name %in% c("Alphapapillomavirus", "Nicoomyvirus", "Myrnavirus")) %>% # filter out known
  filter(genus_tax_id != -200) %>%
  summarize(mean_nt_rpm = mean(nt_rpm, na.rm = TRUE),
            total_reads = sum(nt_count, na.rm = TRUE),
            .groups = "drop") %>%
  na.omit() %>%
  arrange(desc(mean_nt_rpm))

# Select top 10 genus names by mean nt_rpm
top_genus_names <- genus_summary %>%
  group_by(genus_name, library_prep) %>%
  summarize(mean_rpm = mean_nt_rpm, .groups = "drop") %>%
  arrange(desc(mean_rpm)) %>%
  slice_head(n = 20) %>%
  pull(genus_name)

# Filter data for top taxa
top_genus_data <- virus_data %>%
  filter(genus_name %in% top_genus_names)
  #na.omit()

```

```

# Create overall table with Top Viral Genera
top_genus_summary = top_genus_data %>%
  drop_na(genus_name) %>%
  group_by(genus_name) %>%
  summarise(mean_rpm = mean(nt_rpm), .groups = "drop")
#slice_head(n=20)

# get frequency of each genus
genus_counts <- virus_data %>%
  drop_na(genus_name) %>%
  count(genus_name, sort = TRUE) %>%
  mutate(genus_name = fct_reorder(genus_name, n))

# get proportion of time genus is detected in each sample
virus_sample_per = virus_data %>%
  drop_na(genus_name) %>%
  group_by(genus_name) %>%
  summarize(
    perc_of_samples = n_distinct(sample_name) / n_distinct(.$sample_name)
  ) %>%
  arrange(desc(perc_of_samples))

# join back to overall table
top_genus_summary = top_genus_summary %>%
  left_join(genus_counts, by = "genus_name") %>%
  left_join(virus_sample_per, by = "genus_name") %>%
  left_join(genus_composition, by = "genus_name") %>%
  arrange(desc(perc_of_samples))

top_genus_summary %>%
  rename(
    "Genus" = genus_name,
    "rPM" = mean_rpm,
    "Species detections" = n,
    "Percent of Samples" = perc_of_samples,
    "Relative Abundance*" = relative_abundance
  ) %>%
  select(-total_abundance) %>%
  kable(
    format = "latex", # or "latex" if using LaTeX
    caption = "Most abundant genera across all samples",
    digits = 2,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )

```

```
top_genus_summary
```

Most abundant genera across all samples

Genus	rPM	Species detections	Percent of Samples	Relative Abundance*
Orthopicobirnavirus	161.09	338	1.00	34.91
Rotavirus	396.57	144	0.91	36.62
Enterovirus	79.83	79	0.85	4.04
Sapovirus	40.18	47	0.83	1.21
Bocaparvovirus	93.34	113	0.78	6.76
Circovirus	34.75	31	0.67	0.69
Pasivirus	21.75	54	0.65	0.75
Kobuvirus	93.38	16	0.35	0.96
Seadornavirus	74.12	19	0.35	0.90
Buhlduvirus	22.56	3	0.07	0.04
Kahnovirus	28.14	1	0.02	0.02

A tibble: 11 x 6

```

genus_name mean_rpm n perc_of_samples total_abundance relative_abundance
1 Orthopicob~ 161. 338 1 54447. 34.9
2 Rotavirus 397. 144 0.913 57106. 36.6
3 Enterovirus 79.8 79 0.848 6307. 4.04
4 Sapovirus 40.2 47 0.826 1889. 1.21
5 Bocaparvov~ 93.3 113 0.783 10547. 6.76
6 Circovirus 34.7 31 0.674 1077. 0.691
7 Pasivirus 21.8 54 0.652 1175. 0.753
8 Kobuvirus 93.4 16 0.348 1494. 0.958
9 Seadornavi~ 74.1 19 0.348 1408. 0.903
10 Buhlduvirus 22.6 3 0.0652 67.7 0.0434
11 Kahnovirus 28.1 1 0.0217 28.1 0.0180

```

Virus Overview

```

# get all unique non phage virus genera across all samples
dim(table(virus_data$genus_name))

```

```
## [1] 189
```

```

# get all unique non phage virus species across all samples
dim(table(virus_data$name))

```

```
## [1] 684
```

Charting the Viral Community

```

library(forcats)
library(RColorBrewer)
library(viridis)

```

```
## Loading required package: viridisLite
```

```
library(scales)
```

```

##
## Attaching package: 'scales'

```

```
## The following object is masked from 'package:viridis':
##
##   viridis_pal

## The following object is masked from 'package:microbiome':
##
##   alpha

## The following object is masked from 'package:purrr':
##
##   discard

## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(ggtext)
library(treemap)
library(treemapify)
```

```
# Count occurrences of each genus (or use your abundance measure)
genus_counts <- virus_data %>%
  drop_na(genus_name) %>%
  count(genus_name, sort = TRUE) %>%
  mutate(genus_name = fct_reorder(genus_name, n))

# For top N genera (adjust N as needed)
top_genera <- genus_counts %>%
  slice_head(n = 20) # Show top 20 genera

# Percent of samples lollipop
top_genera_lollipop <- ggplot(top_genus_summary, aes(x = perc_of_samples, y = reorder(genus_name, perc_of_samples))) +
  geom_segment(aes(x = 0, xend = perc_of_samples, y = genus_name, yend = genus_name),
    color = "gray70", size = 0.5) +
  geom_point(aes(color = perc_of_samples), size = 3, alpha = 0.8) +
  scale_color_viridis_c(name = "Count", option = "turbo", direction = 1) +
  scale_x_continuous(expand = c(0, 0), labels = comma_format()) +
  labs(
    title = "a",
    x = "Proportion (%)",
    y = "Viral Genus"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray60"),
    plot.caption = element_text(size = 10, color = "gray50", hjust = 1),
    axis.title = element_text(size = 12, face = "bold"),
    axis.text.y = element_text(size = 10, face = "italic"),
    axis.text.x = element_text(size = 10),
    legend.position = "none",
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
```

```

    plot.margin = margin(20, 20, 20, 20)
  )

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

# Relative Abundance Treemap
top_genera_treemap <- ggplot(top_genus_summary, aes(area = relative_abundance, fill = relative_abundance)) +
  geom_treemap(alpha = 0.8, color = "white", size = 1) +
  geom_treemap_text(color = "white", place = "centre", size = 10,
                    family = "Arial", fontface = "bold") +
  scale_fill_viridis_c(name = "Relative Abundance (%)", option = "turbo", direction = 1) +
  labs(
    title = "b",
    caption = "Relative abundance shown by area size and color"
  ) +
  theme_void(base_size = 12) +
  theme(
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray60"),
    plot.caption = element_text(size = 10, color = "gray50", hjust = 1),
    legend.position = "right",
    plot.margin = margin(20, 20, 20, 20)
  )

# genus comp by lib prep
top_genus_composition_prep <- taxa_table_genus %>%
  group_by(genus_name, library_prep) %>%
  rename(
    "Library Prep" = library_prep
  ) %>%
  summarize(mean_rpm = mean(nt_rpm, na.rm = TRUE), .groups = 'drop') %>%
  add_count(genus_name) %>%
  filter(n == 2) %>%
  group_by(genus_name) %>%
  mutate(total = mean(mean_rpm)) %>%
  ungroup() %>%
  slice_max(total, n = 20)

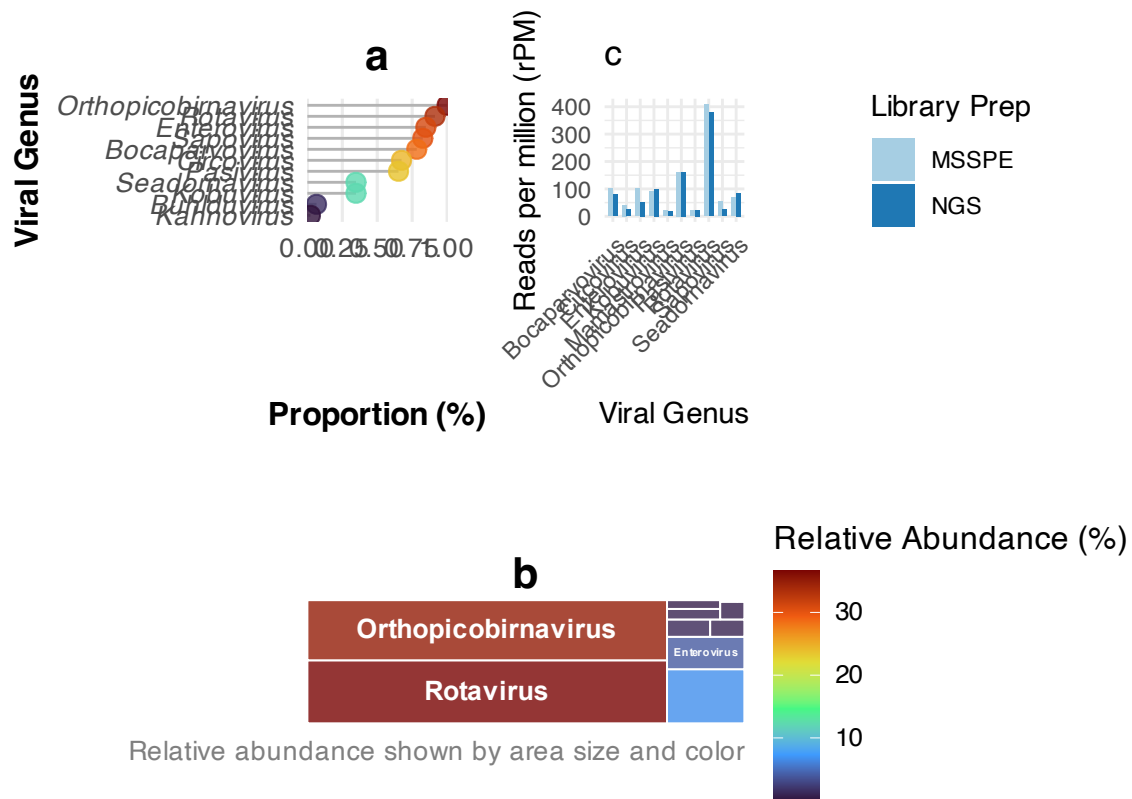
#4 Create a boxplot for the top 10 taxa, grouped by control status
genus_boxplot <- ggplot(top_genus_composition_prep, aes(x = genus_name, y = mean_rpm, fill = `Library Prep`)) +
  geom_bar(stat = "identity", position = "dodge", width = 0.7) +
  scale_fill_brewer(palette = "Paired") +
  labs(title = "c",
    x = "Viral Genus",
    y = "Reads per million (rPM)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

combined_top_genera_plots = (top_genera_lollipop | genus_boxplot) / top_genera_treemap +

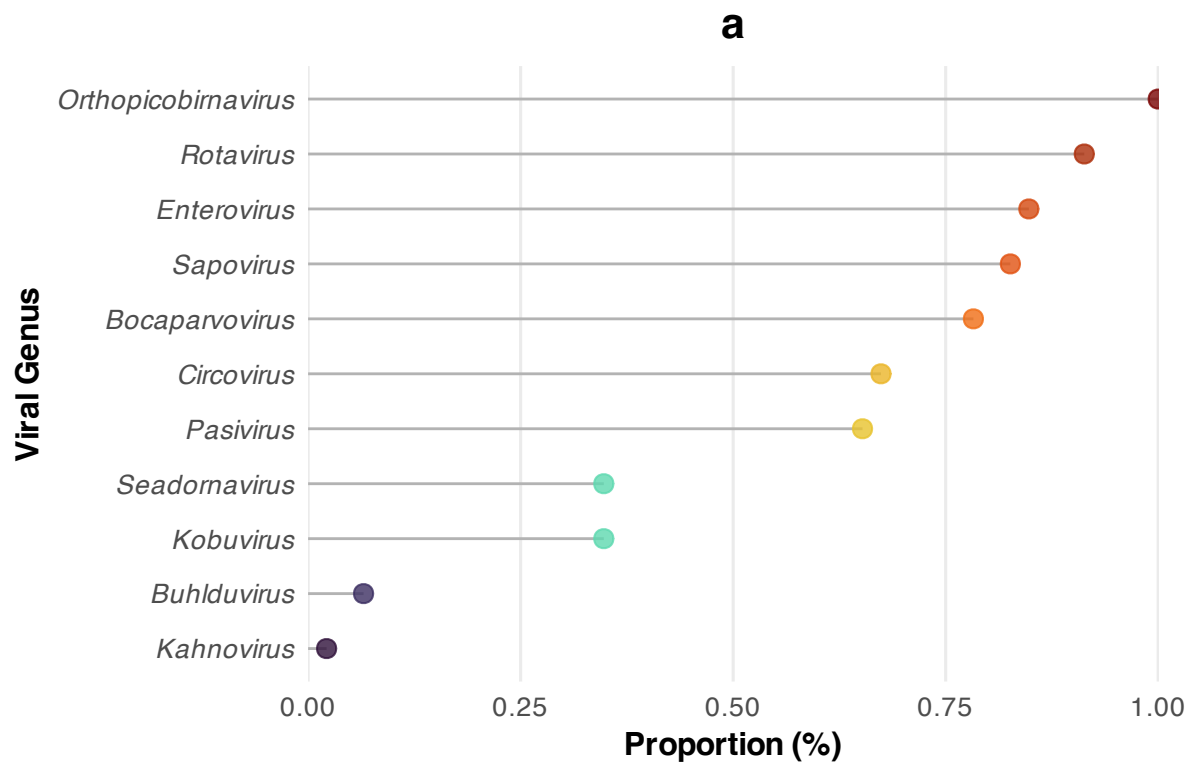
```

```
plot_layout(ncol = 1, heights = c(6.5, 6.5))
```

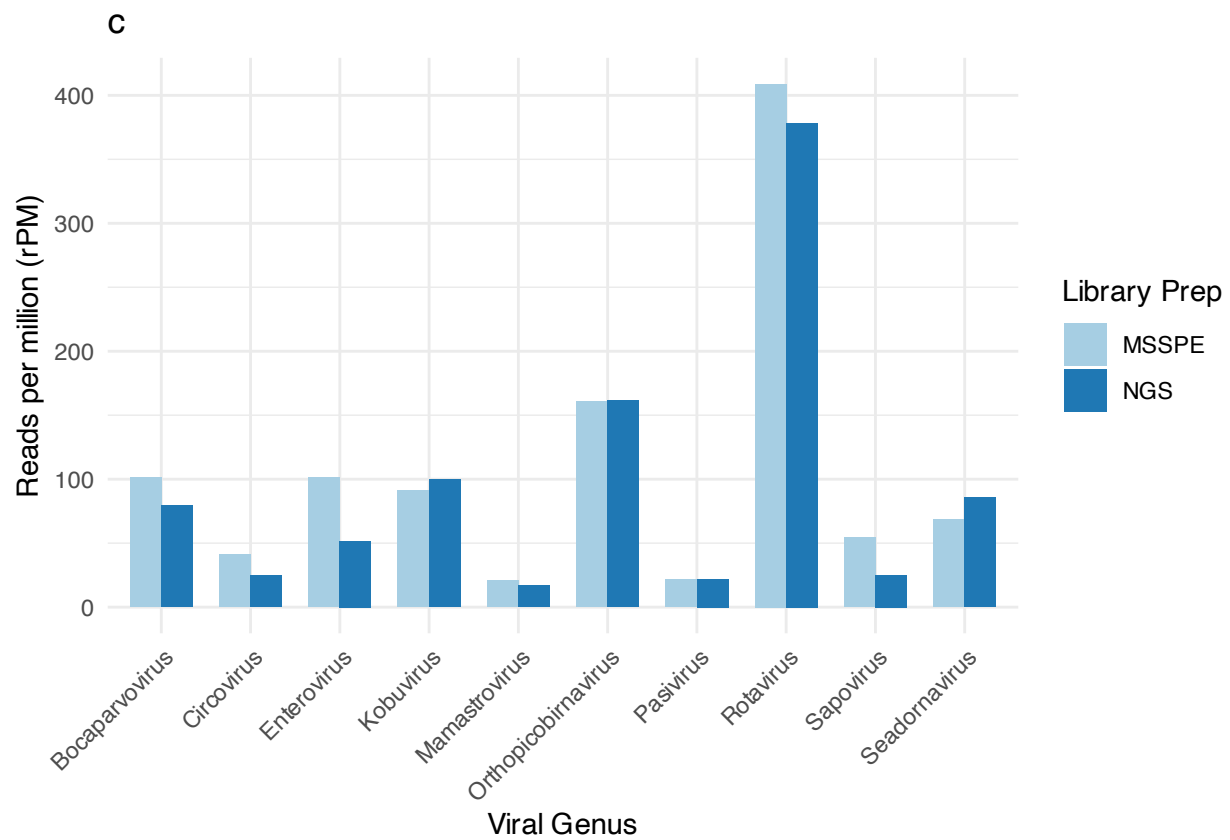
```
combined_top_genera_plots
```



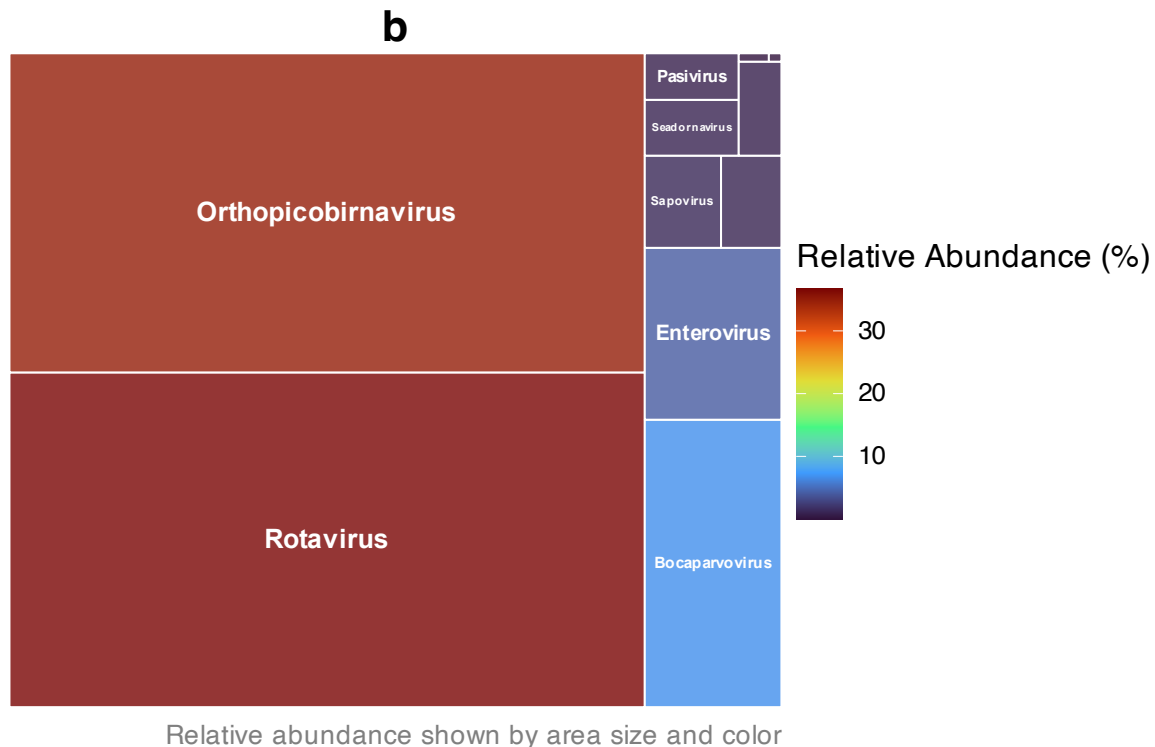
```
top_genera_lollipop
```



genus_boxplot



top_genera_treemap



```
# Summarize at different taxonomic levels for each sample
phylum_composition_samples <- taxa_table %>%
  group_by(category, base_sample) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(base_sample) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)
```

`summarise()` has grouped output by 'category'. You can override using the
`groups` argument.

```
# Visualize composition by sample
tax_comp_plot_all_samples = ggplot(phylum_composition_samples, aes(x = base_sample, y = relative_abundance)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "Taxonomic Composition across All Samples",
       x = "Sample",
       y = "Relative Abundance (%)",
       fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
tax_comp_plot_all_samples
```

